

CSE400 – Fundamentals of Probability in Computing

Lecture 7: Expectation, CDFs, PDFs and Problem Solving

Lecture Scribe

1 Overview of the Lecture

This lecture discusses important concepts used in probability theory and statistical modeling:

1. Cumulative Distribution Function (CDF)
2. Probability Density Function (PDF)
3. Expectation of Random Variables
4. Expectation of a Function of a Random Variable
5. Linear Operations with Expectation
6. Moments and Central Moments
7. Variance, Skewness and Kurtosis

These concepts help describe the behavior, average value, and distribution of random variables.

2 Cumulative Distribution Function (CDF)

2.1 Definition

The **Cumulative Distribution Function (CDF)** of a random variable X is defined as

$$F_X(x) = P(X \leq x)$$

It represents the probability that the random variable takes a value less than or equal to x .

2.2 Properties of the CDF

1. Non-Decreasing Property

$$F_X(x_1) \leq F_X(x_2) \quad \text{if } x_1 < x_2$$

2. Limits at Infinity

$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow \infty} F_X(x) = 1$$

3. Right Continuity

$$\lim_{h \rightarrow 0^+} F_X(x + h) = F_X(x)$$

4. Probability Between Two Values

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

3 Probability Density Function (PDF)

3.1 Definition

For a continuous random variable, the **Probability Density Function (PDF)** is defined as

$$f_X(x) = \frac{d}{dx} F_X(x)$$

3.2 Relationship Between PDF and CDF

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

Thus,

- PDF is the derivative of the CDF
- CDF is the integral of the PDF

3.3 Properties of PDF

1. Non-negativity

$$f_X(x) \geq 0$$

2. Total Probability Equals 1

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

3. Probability Between Two Points

$$P(a < X < b) = \int_a^b f_X(x) dx$$

4 Expectation of Random Variables

4.1 Definition

The **Expectation** (Expected Value) represents the average value of a random variable over many repeated experiments.

4.2 Expectation of Discrete Random Variable

$$E[X] = \sum x_i p_i$$

4.3 Expectation of Continuous Random Variable

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

5 Expectation of a Function of a Random Variable

Sometimes we compute the expectation of a function $g(X)$.

5.1 Discrete Case

$$E[g(X)] = \sum g(x_i) P(X = x_i)$$

5.2 Continuous Case

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

6 Linearity of Expectation

For constants a and b :

$$E[aX + b] = aE[X] + b$$

For two random variables X and Y :

$$E[aX + bY] = aE[X] + bE[Y]$$

Linearity holds even when the random variables are not independent.

7 Moments of Random Variables

The n th moment about the origin is defined as

$$E[X^n]$$

Moments describe the mean, spread, and shape of a distribution.

8 Central Moments

Central moments are calculated about the mean:

$$E[(X - \mu)^n]$$

where

$$\mu = E[X]$$

9 Variance

Variance measures the spread of the distribution.

$$Var(X) = E[(X - \mu)^2]$$

Equivalent form:

$$Var(X) = E[X^2] - (E[X])^2$$

10 Skewness

Skewness measures the asymmetry of a distribution.

- Positive Skewness: Tail extends to the right.
- Negative Skewness: Tail extends to the left.

11 Kurtosis

Kurtosis measures the peakedness and heaviness of the tails of a distribution.

- High kurtosis: heavy tails and sharp peak
- Low kurtosis: flatter distribution

12 Summary

Key concepts covered in this lecture include:

CDF

$$F_X(x) = P(X \leq x)$$

PDF

$$f_X(x) = \frac{d}{dx}F_X(x)$$

Expectation

$$E[X] = \sum x_i p_i$$

or

$$E[X] = \int x f_X(x) dx$$

Additional topics include expectation of functions, linearity of expectation, moments, variance, skewness, and kurtosis.